HAIR LOSS
02

Minoxidil

Pyrimidine-derived vasodilator: scalp solution or foam, and tablets for high blood pressure



CLASS
Vasodilator

STUDIED FOR
Hair loss, high blood pressure

HUMAN EVIDENCE
Large trials; FDA approved

US STATUS
FDA approved (Rogaine OTC); oral hair off-label

STUDIED FOR

Male pattern loss	High blood pressure	Oral for hair loss
HUMAN	HUMAN	HUMAN

Oral use for hair is off-label. In a 24-week trial of 90 men, oral minoxidil was not superior to topical on hair density.

HOW IT WORKS IN THE BODY

Minoxidil ▶ Small arteries ▶ Vasodilation ▶ BP, hair growth

- **Direct-acting peripheral vasodilator,** per the tablet label
- **Lowers blood pressure** by reducing peripheral vascular resistance
- **Extra body hair** seen in about 80% of people on the tablets
- **For hair growth,** the mechanism is not established

EVIDENCE & WHAT'S KNOWN

EVIDENCE

FDA APPROVED
LEVEL 5 OF 5

- Topical approved in 1988 (Rogaine)
- 48-week RCT: 5% vs 2% vs placebo, 393 men
- Oral for hair: one RCT, small open studies

NOT YET ESTABLISHED

- Long-term oral use for hair
- Compounded combination sprays
- Animal heart findings in people

QUICK PRACTICAL INFO

HALF-LIFE
4.2 hours on average (tablets)

ROUTES STUDIED
On the scalp (solution, foam); by mouth (tablets)

PHARMACOKINETICS
Tablets: peak within 1 hour; about 90% metabolized (glucuronidation)

SOLD AS
OTC scalp solution or foam (Rogaine); Rx tablets

MONITORED IN TRIALS
Hair counts, scalp irritation, heart symptoms, swelling

SPORT (WADA)
Not on the 2026 list

REGULATORY

FDA	Topical OTC (1988, 1997); tablets for high BP
FDA	Oral use for hair loss is off-label
EMA	Topical authorised nationally in EU states
WADA	Not on the 2026 list

REFERENCES

1. FDA, minoxidil tablets label (DailyMed), Aug 2026
2. Olsen et al., J Am Acad Dermatol, 5% vs 2% topical minoxidil, Sep 2002
3. Penha et al., JAMA Dermatology, oral vs topical minoxidil, Jun 2024

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